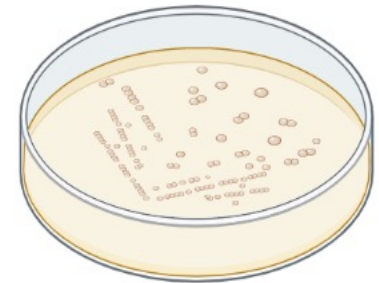


Dilution and Plating

LaBella Lab
Wild Yeast Project

Outline

- Lab notebooks and entries
- Protocol review
- Preparing notebooks



Created using 

What is a Lab Notebook?

- A research record that documents the entire research process:
 - formulating a question or hypothesis
 - designing experiments and studies
 - developing research protocols
 - generating, analyzing, and interpreting data
- A notebook provides sufficient detail so that the research can be reproduced.
 - A scientifically literate person should be able to use the documentation provided in the notebook to navigate rationale, methods, analysis, results, and basis for the major conclusions
- A legal document that may be used as evidence.
 - e.g., to support a patent application, resolve a patent dispute, to defend research against accusations of misconduct
- The notebook represents a scientific legacy of your research group.



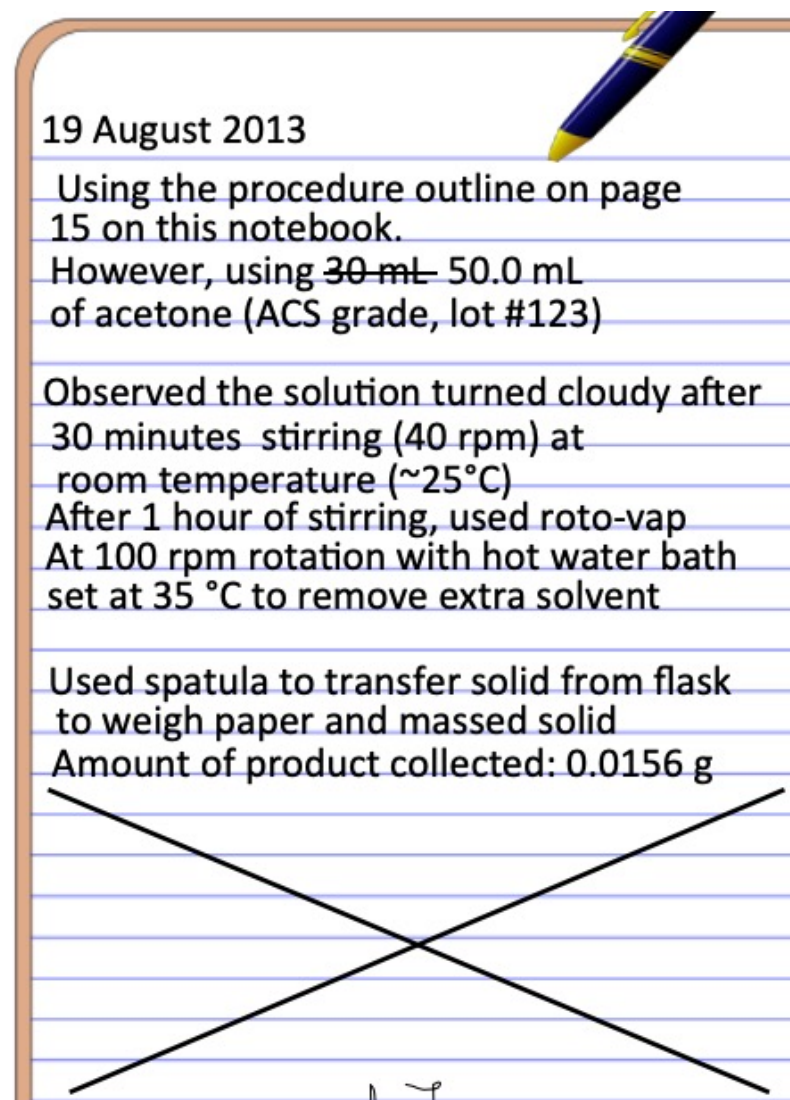
What is a Lab Notebook.... *not*?

- Personal property of the researcher
 - Your lab notebook is a record of work done using federal grant money and must stay with the LaBella Lab
- A journal
 - Research records document the entire research process
 - Record facts (e.g., no reaction occurred under these conditions), not opinions (e.g., this reaction will never work)
- A record of communications within a research group



Good Notebook-keeping

- Use an ink pen
- Write as you go:
 - WHEN – Date (YYYY-MM-DD)
 - HOW – Protocol
 - Include reagents, solutions, type of water, etc.
 - WHAT – Observations
 - Everything that happens (planned or unplanned), raw data, reference to external data
- Enter in consecutive page order; X-out unused pages
 - New experiment? New page
- Draw a single line through errors; write correct info adjacently
- **Poor record-keeping can lead to research misconduct allegations**



Columbia University ReaDI Program



Dilution and Plating Protocol

- Step 1: Observations

- What do you observe about each sample?
- How does it look?
 - Are there bubbles? Do you see any “fuzzy” or filamentous yeasts? What color is the sample?
 - Has the substrate changed? What does it look like?
- What does the sample smell like?
 - *Never smell a sample directly*; instead, hold the tube away from your body and use a hand to gently fan or waft the smell towards you.

- Step 2: Dilution

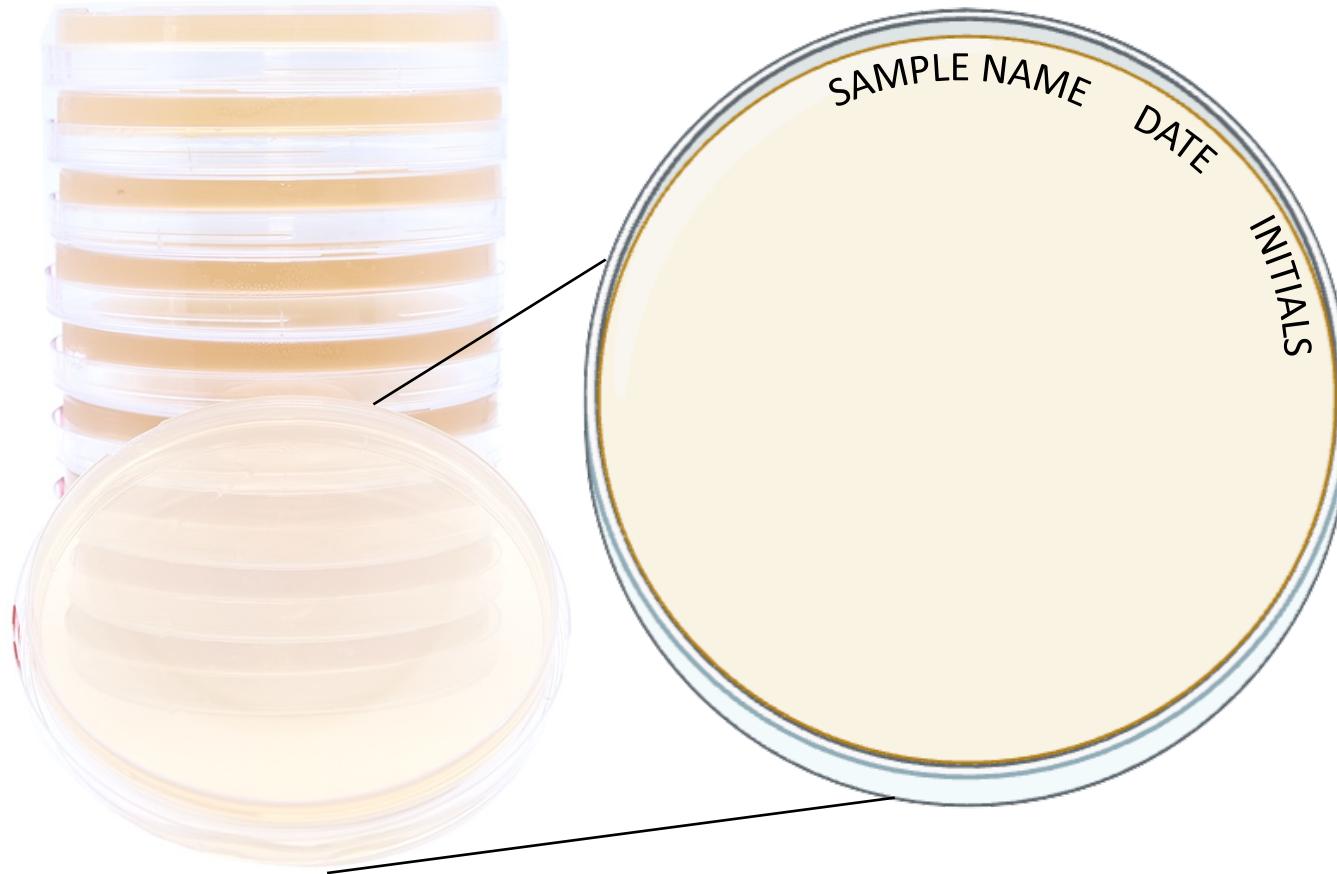
- Perform a serial dilution for each sample (1:10,000 final dilution)
 - Green Dot = 100X dilution, black dot = 10,000X dilution

- Step 3: Plating

- Label one YPD (Yeast extract, Peptone, Dextrose) agar plate per sample
- Plate diluted culture using glass beads



Plate Labeling



- Write on the edge of the bottom of the plate in permanent marker
- Include the following:
 - Sample Name
 - AW_PT_01
 - Date Plated
 - 6/10/2026
 - Initials of Plater
 - AKW

Notebook Prep

Step 1 - Sample Observation

For each of your samples, observe the following details and report the results. *Be sure to leave room in your notebook between the questions*

1. **Bubbles:** Are there any bubbles in your sample? This indicates the presence of fermentation and likely yeast.
2. **Fuzzy Fungi:** Are there any "fuzzy" looking fungi in your sample? These are likely filamentous fungi.
3. **Color:** What color is your sample?
4. **Substrate:** What does your substrate look like now?
5. **Smell:** Do NOT smell your sample directly. Hold the tube slightly away from your body, then use your hand to gently wave or fan the smell toward your nose. What does your sample smell like?

- Write out observation prompts (see handout; leave room for each of your samples)
- Write down each step of the protocol (see handout)



Lab Work – Groups of 3

Confirm w/ Dr. Ward that you (and your notebook) are ready to go.

While others are in the lab, you may review the protocol, your notes from the previous week, or the papers for journal club at lab meeting.

